

GWAS sample size calculation

June 8, 2026

```
[1]: # *****
# GWAS power calculator using R (version v1.0.0)
#
# Author:
# Patrick Yi Feng Wen
# Lab of Prof. HM Wong,
# Faculty of Dentistry, The University of Hong Kong
# For issues, please contact the PI Prof. HM Wong <wonghmg@hku.hk>
#
# Jun 8, 2026
# *****
# This script estimates the statistical power of GWAS. This is used in the
# "Sample size calculation" section of the proposal of our team.
#
# Arguments:
# @n A single value or vector denoting the sample size with which GWAS power
# is to be calculated.
# @b.alt SNP effect size under the alternative hypothesis.
# @f Minor allele frequency.
# @alpha The level of statistical significance. This is usually set to 5e-8 in
# human GWAS, which is the default value of this function.
#
# Output:
# The estimated statistical power of GWAS given the sample size, heritability,
# and level of statistical significance.

gwasPower <- function(n, b.alt, f, alpha = 5e-8) {

  # Standard deviation (SD) of the error term after SNP effect is
  ↪ accounted for
  sigma = sqrt(1 - 2 * f * (1-f) * b.alt^2)

  # Standard error(s) (SE(s)) of the regression coefficient
  ses = sigma/sqrt(n * 2 * f * (1-f))

  # Threshold of chi-square distribution at specified level of
  ↪ significance
```

```

q.thresh = qchisq(alpha, df = 1, ncp = 0, lower = F)

# Statistical power returned
power = pchisq(q.thresh, df = 1, ncp = (b.alt/ses)^2, lower=F)

return(power)
}

```

1 Simulating statistical power of GWAS for quantitative traits

```

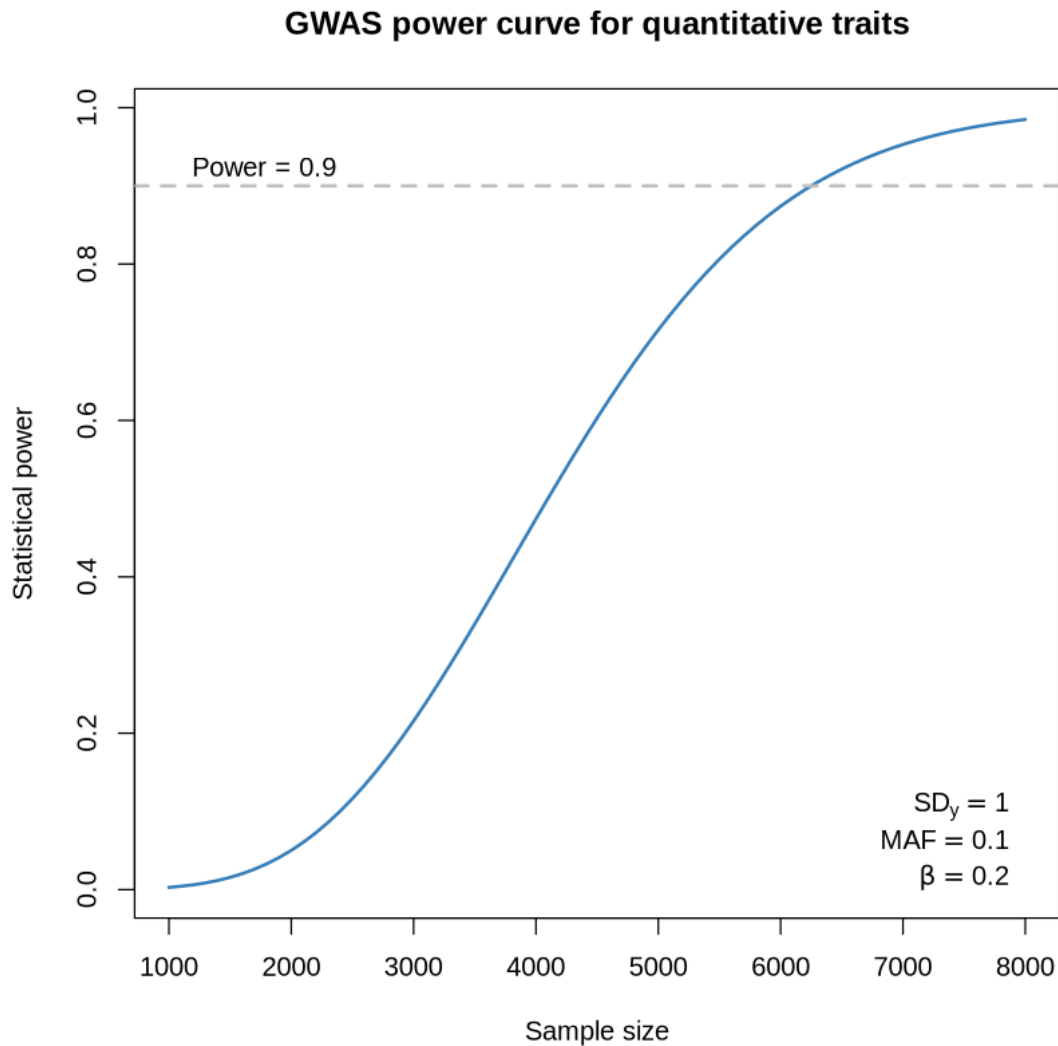
[2]: ns = seq(1000, 8000, 100)
      b.alt <- 0.2
      f <- 0.1
      pwr <- gwasPower(ns, b.alt, f)

```

```

[3]: # Create the base plot
      plot(ns, pwr, col = "#377eb8", xlab = "Sample size", ylab = "Statistical power",
            main = paste0("GWAS power curve for quantitative traits"), type = "l", lwd = 2)
      # Add horizontal line at 90% power
      abline(h = 0.9, lty = 2, col = "grey", lwd = 2)
      # Add text annotations in the lower-right corner
      text(x = max(ns) * 1, y = 0.10,
            labels = bquote(SD[y] == 1), pos = 2, col = "black")
      text(x = max(ns) * 1, y = 0.06,
            labels = bquote(MAF == .(f)), pos = 2, col = "black")
      text(x = max(ns) * 1, y = 0.01,
            labels = bquote(beta == .(b.alt)), pos = 2, col = "black")
      text(x = min(ns) + 1500, y = 0.92,
            labels = "Power = 0.9", pos = 2, col = "black")

```



2 Sample size required to achieve 90% statistical power

```
[4]: ns = 6251
      b.alt <- 0.2
      f <- 0.1
      pwr2 <- gwasPower(ns, b.alt, f)
      pwr2
```

0.900028293618225

At least 6251 individuals are required to achieve a power of 90% when effect size under the alternative hypothesis is assumed 0.2, minor allele frequency is 0.1, and the level of statistical significance is 5×10^{-8} .